Task 1:-

(1). Input

```
1. #include<iostream>
2. using namespace std;
3.    int main( )
4.    {
5.       char letter;
6.       letter = 'A';
7.    cout << letter << \n;
8.          letter = 'B';
9.       cout << letter << \n;
10.          cin >> letter;
11.          return 0;
12.       }
```

(2). Output

```
1. #include <iostream>
2. using namespace std;
3. int main ( )
4. {
5.    char letter ;
6.    letter = 65;
7.    cout << letter << \n;
8.    letter = 66 ;
9.    cout << letter << \n;
10.   Cin >> letter;
11.   return 0;
12.   }
```

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output

A

B

Task 1 : 3

Input

```
1. #include <iostream>
2. using namespace std;
3.    int main( )
4.    {
5.        int letter;
6.        letter = 'A';
7. cout << letter << \n;
8.        letter = 'B';
9.    cout << letter << '\n';
10.       Cin >> letter;
11.       return 0;
12.   }
```

output

65

66

output

A

B

Task 2 :

Input

```
1. #include <iostream>
2. using namespace std;
3. int main()
4. {
5.  int num1, num2;
6. cout << "enter two integers";
7.  Cin >> num1 >> num2;
8. cout << "addition" = << num1+num2
        << endl;
9. cout << "subtraction" = << num1-num2
        << endl;
10. cout << multiplication" = << num1*num2
        << endl;
11. cout << "division" = num1/num2
        << endl;
12. return 0;
13. }
```

out Put

addition = 156

Subtraction = -22

multiplication = 5963

division = 0

∵ int a = 67
int b = 89

Task 3:

Input

```
1. #include <iostream>
2. using namespace std;
3. int main() {
4. int val1, val2, val3;
5. cout << "Please enter your 3 numbers";
6. Cin >> val1 >> val2 >> val3;
7. cout << val1 << "/n" << val2 << "/n" << val3 << "/n";
8. cout << val3 << "/n" << val2 << "/n" << val1 << "/n";
9. System ("PAUSE");
10. return 0;
11. }
```

output

```
Please enter your 3 numbers
45
78
90
sh: PAUSE: not found
45/n78/n90/n90/n78/n49/n.
```

Task 4:

```
1. #include <iostream>
2. #include <cmath>
3. using namespace std;
4. int main ( )
5. {
6.   double x1, y1, x2, y2, d;
7.   cout << "enter x1 and y1:";
8.       cin >> x1 >> y1
9.   cout << "enter x2 and y2
10.      cin >> x2 >> y2 ;
11.  d = sqrt (pow(x2 - x1, 2) + Pow(y2 - y1, 2));
12.      cout << "Distance =" << d << endl;
13.      return 0;
14.  }
```

Output

```
Enter x1 and y1 : 2  3
Enter x2 and y2 : 2  5
   Distance = 5
```

Task 5

(1)

input

```
1.  #include <iostream>
2.  using namespace std;
3.  int main ( )
4.  {
5.      double radius, area;
6.  const double PI = 3.14159;
7.      cout << " enter radius:";
8.      cin >> radius;
9.  area = PI * radius * radius;
10. cout << "Area of circle =" << area << endl;
11.         return 0;
12.         }
```

output

```
Enter radius = 5
Area of circle = 78.5397
```

(2)

input

```
1.  #include <iostream>
2.  using namespace std;
3.  int main( )
4.  {
```

```
5. double length, width, area;
6. cout << " enter length: ";
7.     cin >> length;
8. cout << " enter width: ";
9.     cin >> width;
10. area = length * width;
11. cout << "Area of rectangle=" << area << endl;
12.     return 0;
13  }
```

output

```
enter length: 10
enter width: 5
Area of rectangle = 50.
```

(3)

input

```
1. #include <iostream>
2. using namespace std;
3. int main()
4. {
5. double base, height, area;
6. cout << " enter base: ";
7.     cin >> base;
8. cout << " enter height: ";
9.     cin >> height;
10. area = base * height * 0.5;
11. cout << " area of triangle= " << area << endl;
12.     return 0;
```

out put

```
enter base = 8
enter height = 6
area of triangle = 24
```

Task 6

```
1. #include <iostream>
2. using namespace std;
3. int main ( )
4.  double item1 = 12.95;
5.  double item2 = 24.95;
6.  double item3 = 6.95;
7.  double item4 = 14.95;
8.  double item5 = 3.95;
9.  double subtotal, tax, total;
10.  subtotal = item1 + item2 + item3, item4 + item5;
11.  tax = subtotal * 0.06;
12.  total = subtotal + tax;
13. cout << "price of item 1 : $" << item1 << endl;
14. cout << "price of item2 : $" << item2 << endl;
15. cout << "Price of item 3 : $" << item3 << endl;
16. cout << "Price of item 4 : $" << item4 << endl;
17. cout << "Price of item5 : $" << item5 << endl;
18. cout << "subtotal : $" << subtotal << endl;
19. cout << "sales tax (6%) : $" << tax << endl;
20. cout << "Total purchase : $" << total << endl;
21. ~~cout <<~~ return 0;
22. }
```

Output

```
Price of item 1 : $12.95
Price of item 2 : $ 24.95
Price of item 3 : $ 6.95
Price of item 4 : $ 14.95
Price of item 5 : $ 3.95
   Subtotal : $63.75
Sales tax (6%) = $ 3.825
Total purchase : $67.575
```

Home Task

```
1. # include <iostream>
2. using namespace std;
3. int main ( )
4. {
5.     int num;
6. cout << "Enter a five-digit number: " ;
7.     cin >> num;
8. cout << num / 1000 << " ";
9.   cout << (num / 1000) % 10 << " " ;
10.  cout << (num / 100) % 10 << " " ;
11.  cout << (num / 10) % 10 << " " ;
12.  cout << num % 10 << endl;
13.      return 0;
14.  }
```

output

Enter a five digit number: 12345

1 2 3 4 5

Bonus Task

Input (i)

```
1- #include <iostream>
2- using namespace std;
3- int main( ) {
4-     int var1, var2, temp;
5- cout<< "enter first number:";
6-     cin >> var1;
7- cout << "enter second number:";
8-     cin >> var2;
9-     temp = var1;
10-    var1 = var2;
11-    var2 = temp;
12- cout << "After swapping:" << endl;
13- cout << "var1=" << var1 << endl;
14- cout << "var2:" << var2 << endl;
15-     return 0;
16- }
```

Output:

Enter first number: 10
Enter second number: 20
After swapping;
var 1 = 20
var 2 = 10

(2)

Input

```
1- #include <iostream>
2- using namespace std;
3- int main( ) {
4-    char ch;
5-    cout << "Enter a lowercase alphabet: ";
6-       cin >> ch;
7-       ch = ch - 32;
8.    cout << "Uppercase alphabet: " << ch << endl;
9-       return 0;
10-   }
```

Output

Enter a lowercase alphabet: a
Uppercase alphabet : A

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